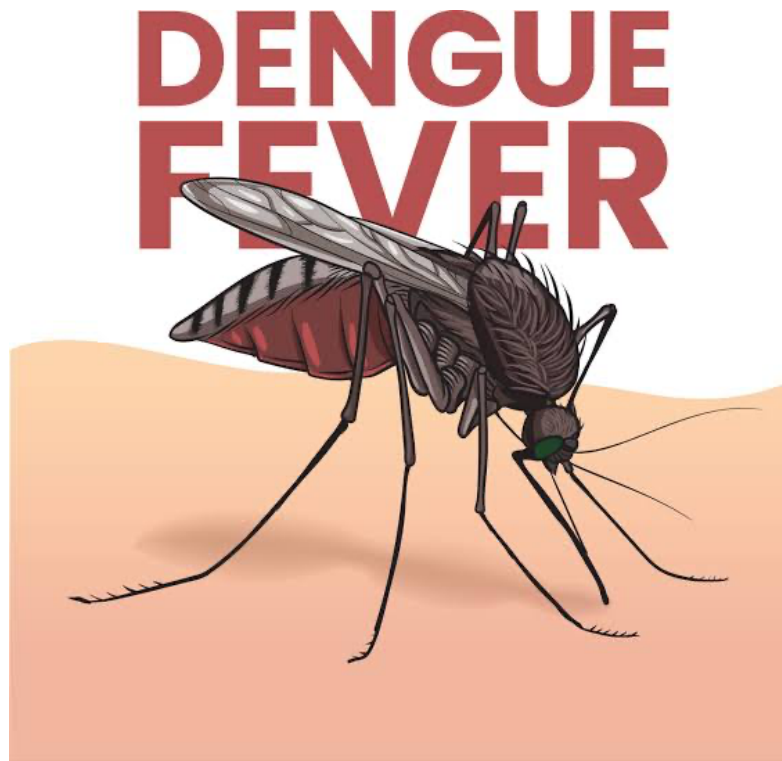


DengAI: predicting weekly dengue cases

Goal: estimate reported dengue cases each week in San Juan and Iquitos using environmental measurements

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The problem: dengue outbreaks are hard to anticipate



Dengue is a mosquito-borne virus spread by mosquitoes in tropical and subtropical regions. Severe cases can be fatal.

Outbreaks follow the weather. Warmth, humidity and rainfall drive mosquito breeding, so case counts rise and fall with the climate, often weeks later.

Why it matters

Earlier warning: predict a surge before it shows up in clinics

Lower cost: target spraying and breeding-site clean-up where and when it is needed

Better care: prepare beds, staff and test kits ahead of peaks

Our ML model: learn the link between environmental conditions and case counts, then forecast weekly cases for each city.

A supervised regression problem

We predict weekly dengue case counts for San Juan (Puerto Rico) and Iquitos (Peru).

```
In data.shape, labels.shape, test.shape
```

```
Out ((1456, 24), (1456, 4), (416, 24))
```

1,456 historical weeks, 416 future weeks, and 24 input columns.

```
In labels.head()
```

Out	city	year	weekofyear	total_cases
0	sj	1990	18	4
1	sj	1990	19	5
2	sj	1990	20	4
3	sj	1990	21	3
4	sj	1990	22	6

total_cases is the value to predict:
a weekly count of reported cases.

For example, the first San Juan week has
4 cases.

The environmental features come from four sources

Group	Column names
Location and time	city, year, weekofyear, week_start_date
1. Weather stations	station_avg_temp_c, station_min_temp_c, station_max_temp_c, station_diur_temp_rng_c, station_precip_mm
2. Satellite rainfall	precipitation_amt_mm
3. Climate reanalysis	reanalysis_air_temp_k, reanalysis_avg_temp_k, reanalysis_min_air_temp_k, reanalysis_max_air_temp_k, reanalysis_dew_point_temp_k, reanalysis_tdtr_k, reanalysis_relative_humidity_percent, reanalysis_specific_humidity_g_per_kg, reanalysis_precip_amt_kg_per_m2, reanalysis_sat_precip_amt_mm
4. Satellite vegetation	ndvi_ne (northeast of the city centre) ndvi_nw (northwest), ndvi_se (southeast), ndvi_sw (southwest)

Source: [DrivenData feature definitions](#)

Check the cities and their timelines

San Juan (sj) and Iquitos (iq) have different amounts of history.

In

```
data.groupby("city")["week_start_date"]  
    .agg(["count", "min", "max"])
```

In

```
test.groupby("city")["week_start_date"]  
    .agg(["count", "min", "max"])
```

Out

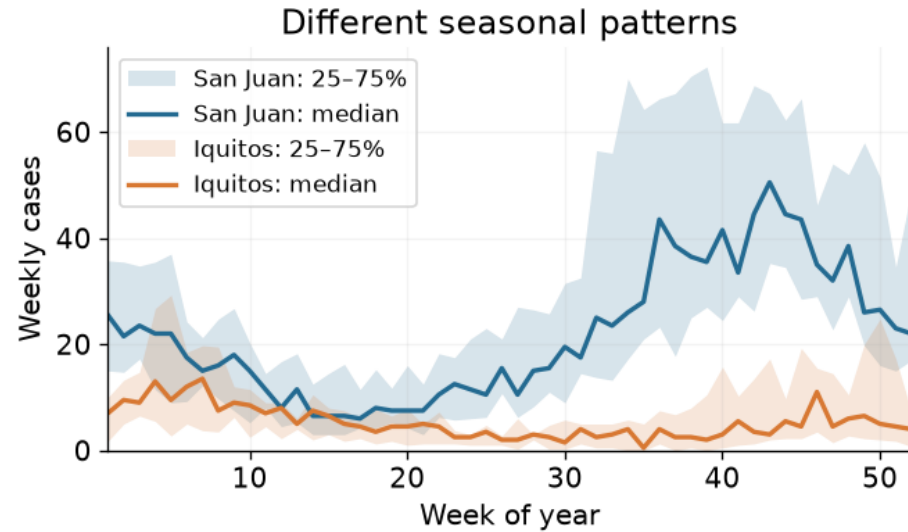
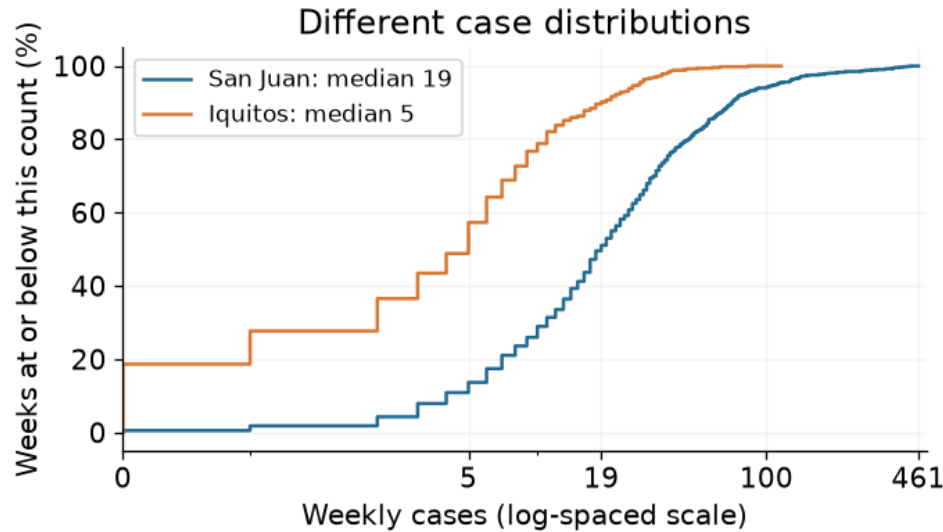
city	count	min	max
iq	520	2000-07-01	2010-06-25
sj	936	1990-04-30	2008-04-22

Out

city	count	min	max
iq	156	2010-07-02	2013-06-25
sj	260	2008-04-29	2013-04-23

Each city's test period follows its training period: about five years for San Juan and three years for Iquitos. We therefore validate on later weeks.

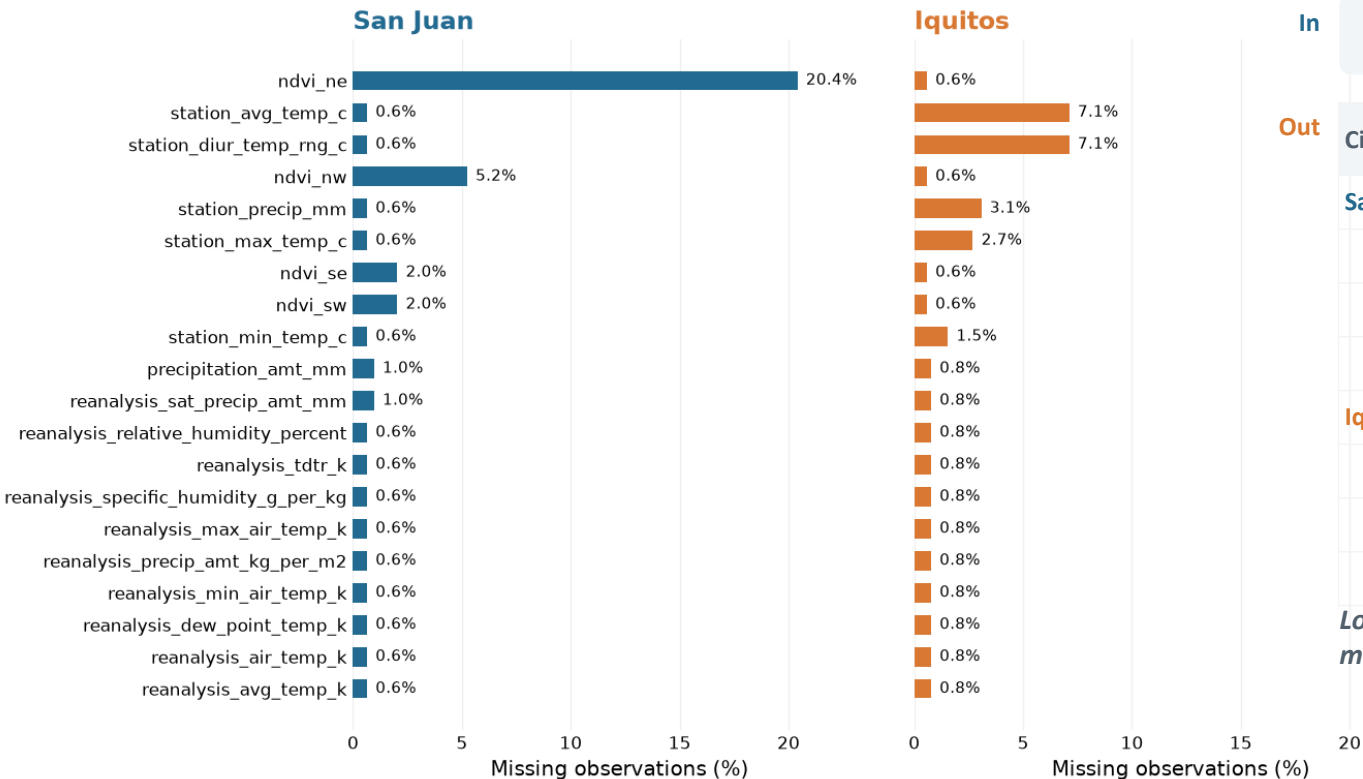
The cities have different case levels and seasons



Decision: perform EDA separately, then prepare and fit a model for each city.

Vegetation has the most missing values

Missing environmental measurements



In

```
ndvi_missing_summary(eda)
```

Out

City	Feature	Missing (%)	Longest gap
San Juan	ndvi_ne	20.4	15
	ndvi_nw	5.2	15
	ndvi_se	2.0	14
	ndvi_sw	2.0	14
Iquitos	ndvi_ne	0.6	1
	ndvi_nw	0.6	1
	ndvi_se	0.6	1
	ndvi_sw	0.6	1

Longest gap = longest run of consecutive missing observations within each city.

How we fill missing measurements

San Juan vegetation

Typical value (median) for the same week of the year, learned from the training data. If unavailable, the overall training median.

Iquitos vegetation

Linear interpolation between the nearest available measurements before and after the gap.

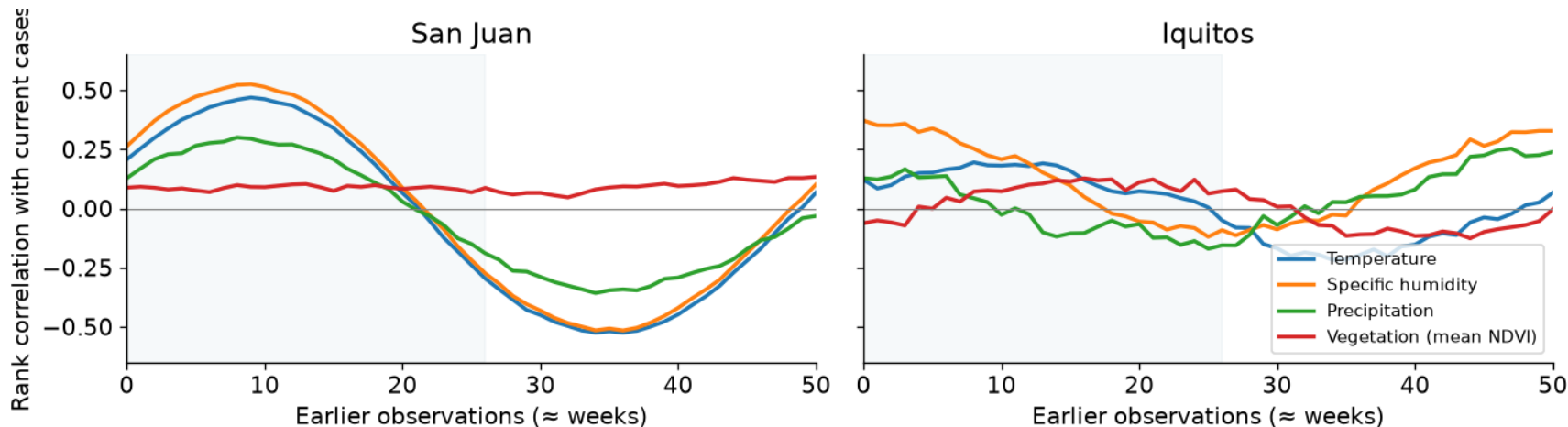
Other weather, both cities

The same interpolation approach for gaps between known values.

Why treat San Juan vegetation differently?

Its gaps run up to 15 consecutive observations. Forward-fill would repeat one old value for months; the same-week median keeps the seasonal shape.

Earlier weather helps explain current cases



The effect is delayed

In San Juan, temperature and humidity correlate most with cases 8–10 weeks earlier

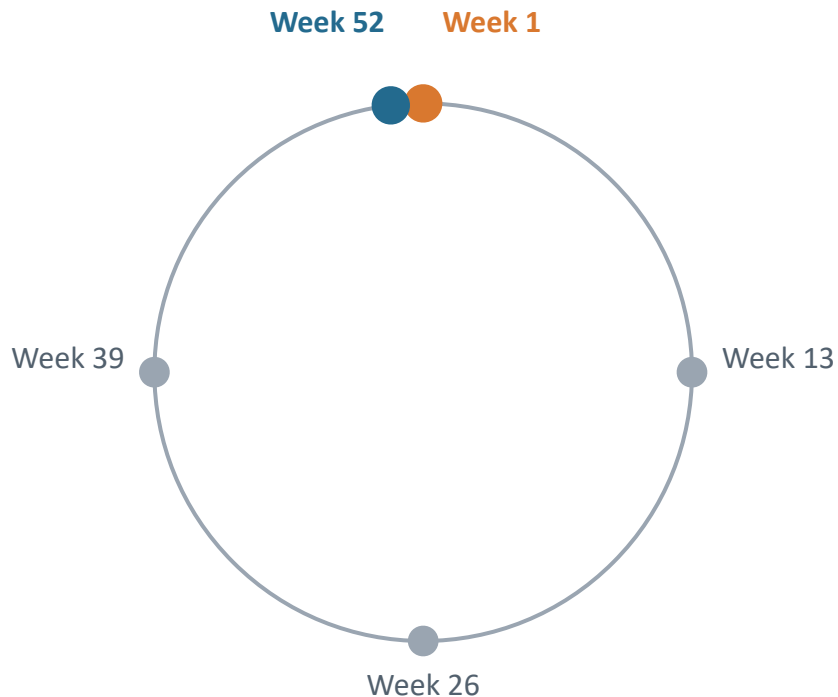
We add lags

Values from 1, 2, 4, 8, 10, 20 and 26 weeks earlier

And rolling summaries

Averages and rainfall totals over the previous 4–26 weeks

Representing the yearly cycle



Weeks 52 and 1 are neighbours, even though their numbers are far apart

We use **cyclical time encoding** for weeks

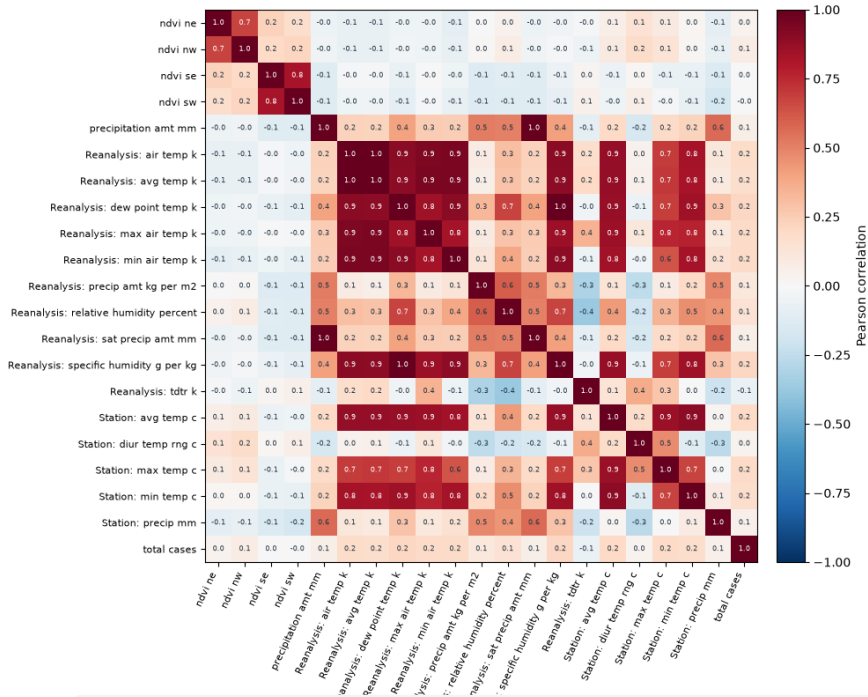
Sine and cosine place each week on a circle

The model sees late December and early January as close together

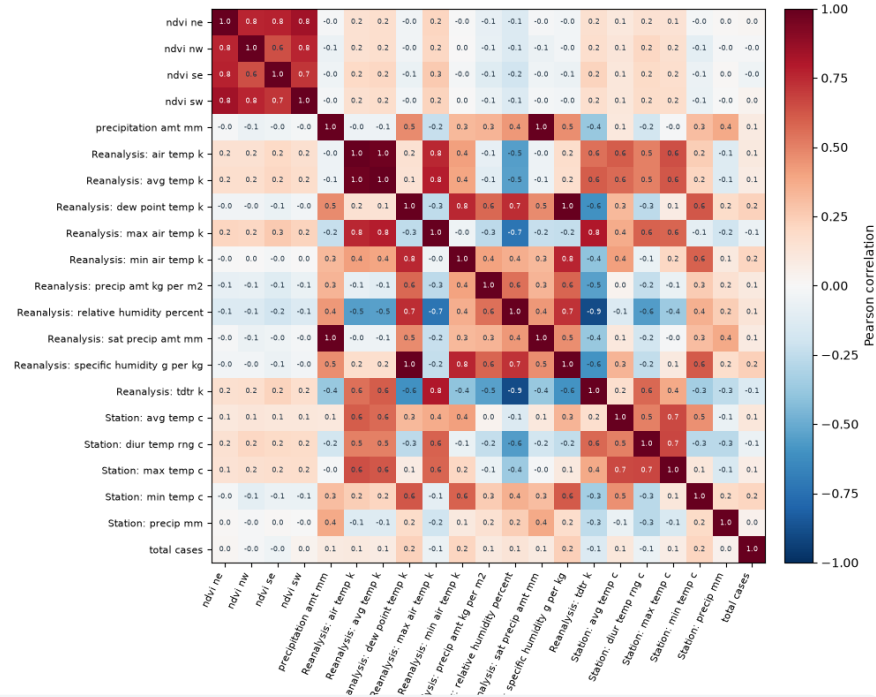
```
sin(2π · week / 52)  
cos(2π · week / 52)
```

Checking for redundant inputs

San Juan - environmental features and weekly cases



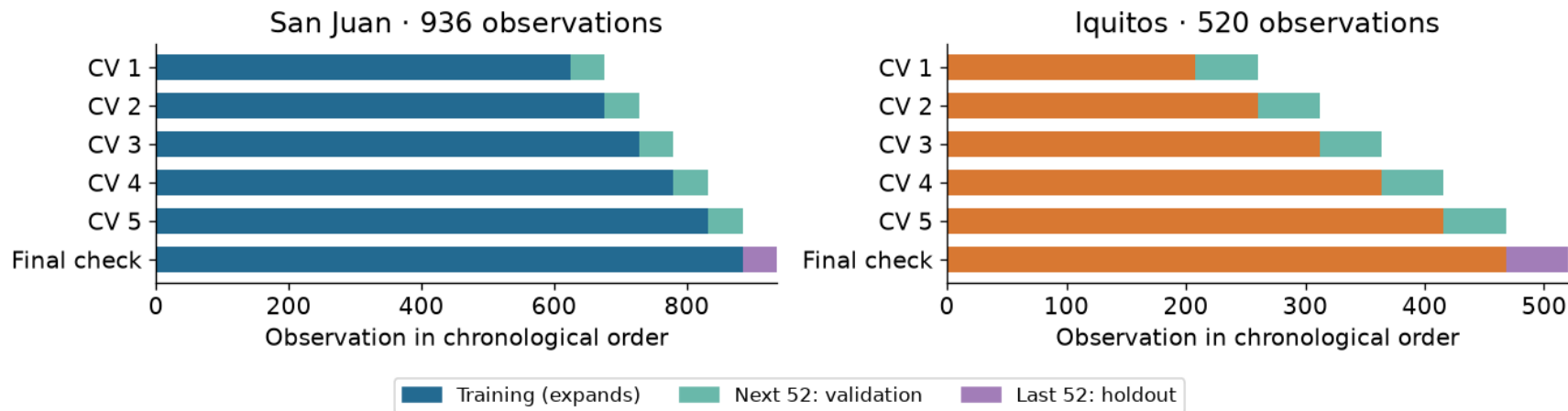
Iquitos - environmental features and weekly cases



Strong correlation suggests overlap, not identity.

Checked directly: **precipitation_amt_mm = reanalysis_sat_precip_amt_mm** in both cities, so the duplicate is **dropped** (along with city, week_start_date and weekofyear).

Time-ordered cross-validation



Keep time order

Train on earlier weeks, validate on later ones

5 folds per city

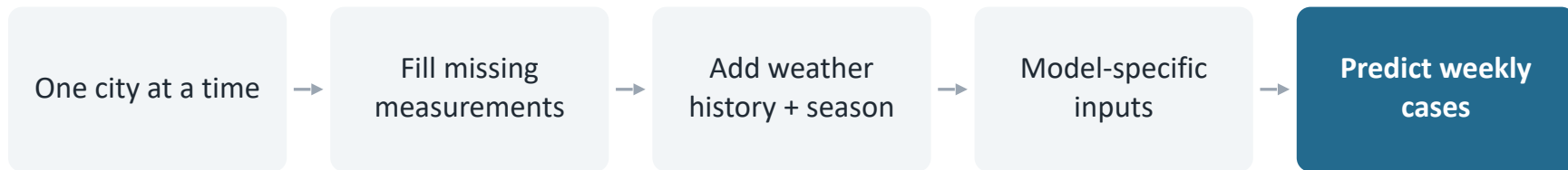
Each predicts the next 52 weeks with more history

Final holdout

Last 52 weeks set aside; choose settings by CV MAE

```
time_cv = TimeSeriesSplit(n_splits=5, test_size=52)
```

One pipeline, refit inside every fold



Avoid leakage

Filling rules, scaling and feature selection are learned from training weeks only

Fair validation

Validation case counts never enter training

Consistent predictions

The same fitted preparation is reused on new data

Feature preparation at a glance

Feature group	San Juan	Iquitos
Vegetation (ndvi_*)	Same-week-of-year training median; overall median as fallback	Interpolate internal gaps
Station, reanalysis and satellite rainfall	Interpolate internal gaps	Interpolate internal gaps
Weather history	Lagged values and rolling summaries	Same
Annual seasonality	Sine/cosine of week of year	Same